

Notes of Introduction of Modern Economic Growth

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1 Dynamic optimization

Acemoglu's textbook¹ used many methods of dynamic optimization when explaining models, therefore, it is necessary to cover this topic in the very first beginning. We will cover the most useful two methods about this, the Lagrangian formulation and the Current-value Hamiltonian, to solve a same sample question.

1.1 The model

Let's look at the model below:

$$\max_{c(t)} \int_0^{\infty} e^{-\rho t} u[a(t), c(t)] dt$$

s.t

$$\dot{a}(t) = f[a(t), c(t)]$$

$a(0)$ is given

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¹Acemoglu, Daron. Introduction to Modern Economic Growth. Princeton ; Oxford: Princeton UP, 2009. Print.

Where $c(t)$ is consumption and we also call it control variable, $a(t)$ is capital and also called state variable, $e^{-\rho t}$ is the discount item and $u(\cdot)$ is the utility function.

1.2 The Lagrangian formulation

To solve this problem at every point of time, a Lagrangian formulation serves as a basic tool:

$$\mathcal{L} = \int_0^\infty e^{-\rho t} \{u(a, c) + q[f(a, c) - \dot{a}]\} dt$$

Where $q(t)$ is a co-state variable and is also known as a shadow value of $a(t)$, in a more general problem, we also use λ .

Now, we solve the Lagrangian formulation:(divide it into two parts)

$$\mathcal{L} = \int_0^\infty e^{-\rho t} [u(a, c) + qf(a, c)] dt - \int_0^\infty (e^{-\rho t} q \dot{a}) dt$$

We first consider a new expression like this:(note that differential after integral primitive function will be equal to itself)

$$\begin{aligned} \int_0^\infty \frac{d}{dt} (e^{-\rho t} q a) dt &= \lim_{t \rightarrow \infty} e^{-\rho t} q(t) a(t) - q(0) a(0) \\ &= \int_0^\infty [e^{-\rho t} \dot{q} a + e^{-\rho t} q \dot{a} + q a (-\rho e^{-\rho t})] dt \\ &= \int_0^\infty (e^{-\rho t} q \dot{a}) dt + \int_0^\infty e^{-\rho t} a (\dot{q} - \rho q) dt \end{aligned}$$

Note that the first item of the last expression(Marked in blue) is exactly the rightmost item of the Lagrangian formulation, therefore:

$$\int_0^\infty (e^{-\rho t} q \dot{a}) dt = - \int_0^\infty e^{-\rho t} a (\dot{q} - \rho q) dt + \lim_{t \rightarrow \infty} e^{-\rho t} q(t) a(t) - q(0) a(0)$$

The Lagrangian formulation now can be re-written as:

$$\mathcal{L} = \int_0^\infty e^{-\rho t} [u(a, c) + qf(a, c)] dt + \int_0^\infty e^{-\rho t} a (\dot{q} - \rho q) dt - \lim_{t \rightarrow \infty} e^{-\rho t} q(t) a(t) + q(0) a(0)$$

FOCs of the Lagrangian formulation are as follows :

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c} = 0 &\implies \frac{\partial u}{\partial c} + q \frac{\partial f}{\partial c} = 0 \\ \frac{\partial \mathcal{L}}{\partial a} = 0 &\implies \frac{\partial u}{\partial a} + q \frac{\partial f}{\partial a} + (\dot{q} - \rho q) = 0 \\ \frac{\partial \mathcal{L}}{\partial q} = 0 &\implies f(a, c) - \dot{a} = 0\end{aligned}$$

1.3 The Current-value Hamiltonian

Then, we use the Current-value Hamiltonian method² to solve this, the current-value Hamiltonian formulation of the same optimization question:

$$\mathcal{H}[a(t), c(t), q(t)] = u[a(t), c(t)] + q(t)f[a(t), c(t)]$$

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial c} = 0 &\implies \frac{\partial u}{\partial c} + q \frac{\partial f}{\partial c} = 0 \\ \frac{\partial \mathcal{H}}{\partial a} = \rho q - \dot{q} &\implies \frac{\partial u}{\partial a} + q \frac{\partial f}{\partial a} = \rho q - \dot{q} \\ \frac{\partial \mathcal{H}}{\partial q} = \dot{a} &\implies f(a, c) - \dot{a} = 0\end{aligned}$$

Transversality condition (T.V.C):

$$\lim_{t \rightarrow \infty} e^{-\rho t} q(t) x(t) = 0$$

We do not do the complex calculation here because we should always bear this in mind: **The results of these two methods are always the same when dealing with the same question.** But, if you are a mathematics lover, you are welcome to prove this by yourself, as I am not such person, my notes do not cover this.

The Current-value Hamiltonian formulation also subjects to a transversality condition, which implies the households will spend all of their wealth and with no debt before they die.

²Note that the Present-value Hamiltonian is very similar to the Current-value one, its formulation is: $\tilde{\mathcal{H}}[a(t), c(t), \phi(t)] = e^{-\rho t} u[a(t), c(t)] + \phi(t) f[a(t), c(t)]$, where $e^{-\rho t}$ is the discount item, $\tilde{\mathcal{H}} = e^{-\rho t} \mathcal{H}$ and $\phi(t) = e^{-\rho t} q(t)$

I expanded the Lagrangian formulation and proved its result as you can see, this is because, for me, and I believe for many students, we are more familiar with Lagrangian than Hamiltonian. The FOCs of Lagrangian are very easy and intuitive, they are equal to zero. While for Hamiltonian, except the FOC wrt control variable is equal to zero, other FOCs are strange and not intuitive.